

Using AI Agents Responsibly in Research

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A full research paper in 24 hours



Can AI conduct research for us?

AI-written papers are weaker in ideas than in execution

AI models rated 912 AI-written economics papers and 41 published papers (Li, 2026)

Execution scored 3.8 vs. 4.4 out of 5, but fewer ideas reached top-journal level (23% vs. 73%)

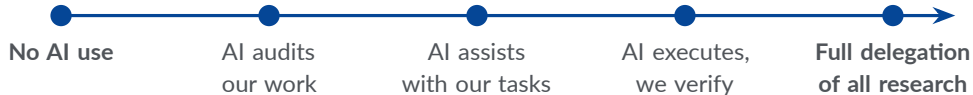
AI agents do the technical work alone, but their papers are rejected

AI agents had six days to answer the question of an unpublished paper (Kirgis et al., 2026)

The paper's authors reviewed the result: the conclusions did not follow from the evidence

In both studies, AI handles the technical work better than the ideas and the research judgment.

The scale of AI use in research



Where on the scale should we use AI for research?

What comes first in AI tooling for research?

Agent-first

Design and optimize autonomous agents



The agent chooses the steps

Test the agent with evals and sandboxes

Skill-first

Design workflows that we or our agents can call



A written workflow sets the steps

Test the workflow with evals and sandboxes

Not mutually exclusive, but this priority shapes how we think about tooling and AI use.

What priority does reproducible research need?

Research must be reproducible



Every run must return the same results

We must be able to verify every step

Agents are probabilistic



The same task can return different results

Agents cannot reliably report their steps

My view: skill-first

How can skill-first help us do reproducible research?

Skill-first allows us to fix the process in a workflow, for us and for our AI agents.

Clear goals and repeatable processes

1

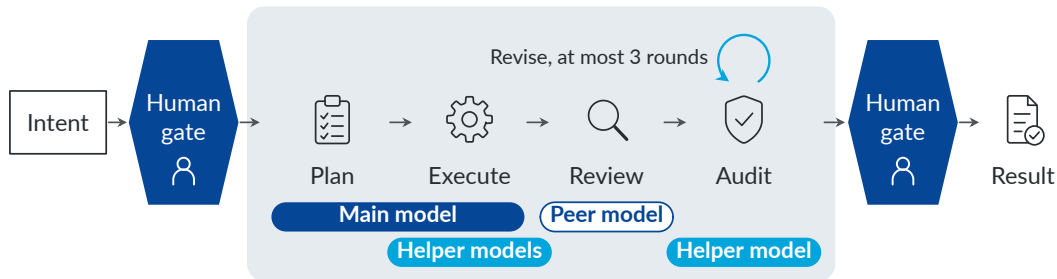
It forces us to set well-defined goals and processes, which is what reproducible research requires.

Clear roles for humans and agents

2

It makes us decide where human judgment is needed and where steps can be delegated to (sub)agents when they don't need verification and can be forgotten.

My current skill template



See the [Anthropic documentation](#) and [agentskills.io](#).

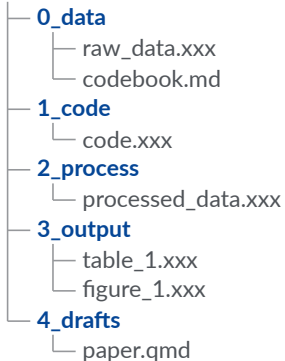
Example: the paper audit skill

Purpose: Does a full audit (mistakes, errors, inconsistencies, and fact-check) in a paper before it goes to a journal. The skill evaluates and reports. I decide what to fix.

- 1 I give it the paper, the appendix, the response letter, and the analysis output
- 2 It lists every table, figure, number, and reference in these files
- 3 It writes a short plan of the errors most likely in this paper
- 4 Four agents each check one part: references, tables, text, and the response letter
- 5 A fresh agent checks each finding against the paper and sets aside what it cannot confirm
- 6 Parts that were missed are checked again, for at most three rounds
- 7 I receive a report that ranks each mistake and gives its page and an exact quote

A reproducible folder structure

Project folder



Four rules

1. Never modify **0_data**
2. Only save data to **2_process**
3. Only load data from **0_data** or **2_process**
4. Anything in **2_process** and **3_output** can be deleted and regenerated from **1_code**

The agent's role in the pipeline

- Agents follow the same rules as human researchers
- Delete all processed data and output: the code rebuilds everything
- **Role 1: agents edit the code, or help us edit it**
 - Already challenging, because AI makes mistakes
 - The researcher stays in the loop
- **Role 2: AI as a source of data**
 - Data an LLM touches becomes generated data, stored as raw data
 - AI never restructures, merges, or formats existing data: that is code's job

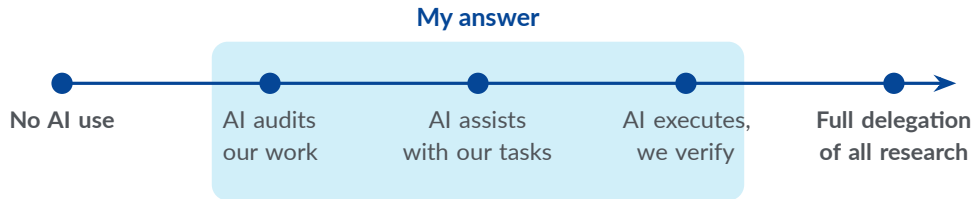
AGENTS.md: rules for every agent

```
# AGENTS.md (excerpt, shortened)
- Never modify the raw data in 0_data/.
- Never edit 2_process/ or 3_output/ by hand.
- Change only 1_code/, 4_drafts/, and the codebook.
- Before a multi-file change, say which files will change.
- Never run git commit or git push.
- Done only when "make clean" then "make" rebuilds everything.
```

- Git enables us to check and audit the changes made by AI agents (researcher-in-the-loop)

Source: github.com/victorvanpelt/research_pipeline_example

My position on the scale



Where on the scale should we use AI for research?

Thank you for your attention

I look forward to hearing your thoughts and experiences.

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- Li, N. (2026). *The ideation bottleneck: Decomposing the quality gap between AI-generated and human economics research*. arXiv. <https://doi.org/10.48550/arXiv.2604.03338>